

Birla Institute of Technology and Science, Pilani
Digital Signal Processing (ECE/EEE F434)

Comprehensive Examination, Open Book, First Semester 2024 – 25

Maximum Time: 60 minutes

04/12/24

Maximum Marks: 30

1. Give a scheme to compute the 3072-point DFT of a sequence of length 3072 using 512-point FFT modules and complex multiplications and additions. Show the scheme in block-diagram form, the design should be cost effective and show complete analysis. For the design, the costs of individual modules are as given below. (15)

- one 512-point FFT module costs Rs. 500
- one multiplier costs Rs. 100
- one delay/advance unit costs Rs. 10
- one adder costs Rs. 50
- one up/down sampler unit costs Rs. 200
- If your design needs any other element, you can specify the same and add the cost as Rs. 100 per element
- the connecting wires do not add to the final cost.

2. If you are given a basic filter module (a hardware or computer subroutine), it is sometimes possible to use it repetitively to implement a new filter with sharper frequency-response characteristics. One approach, as we discussed in the lecture, and you used in the last lab exercise, is to cascade the filter with itself two or more times. Another approach, suggested by Tukey is called “twicing,” as shown in the block diagram given by Figure 1. Note that the overall frequency response is denoted as $G(e^{j\omega})$ (from $x[n]$ to $y[n]$).

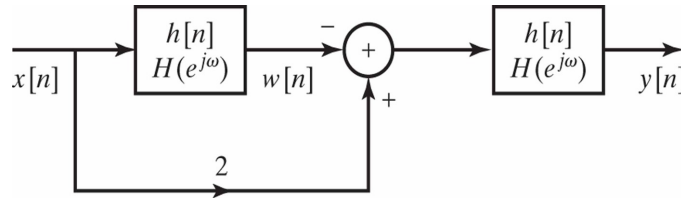


Figure 1: Twicing approach.

- (a) Assume that the basic system has a symmetric finite-duration impulse response, i.e., $h[n] = h[-n]$, $-L \leq n \leq L$. Determine whether the overall response $g[n]$ is (i) FIR and (ii) symmetric. (5)
- (b) Suppose $H(e^{j\omega})$ satisfies following approximation error specifications

$$(1 - \delta_1) \leq H(e^{j\omega}) \leq (1 + \delta_1), \quad 0 \leq \omega \leq \omega_p,$$

$$-\delta_2 \leq H(e^{j\omega}) \leq \delta_2, \quad \omega_s \leq \omega \leq \pi.$$

If the basic system has these specifications, it can be shown that the overall frequency response $G(e^{j\omega})$ (from $x[n]$ to $y[n]$) satisfies the specifications of the

form which are given as

$$\begin{aligned} A &\leq G(e^{jw}) \leq B, \quad 0 \leq w \leq w_p, \\ C &\leq G(e^{jw}) \leq D, \quad w_s \leq w \leq \pi. \end{aligned}$$

Determine A, B, C, and D in terms of δ_1 and δ_2 . If $\delta_1 \ll 1$ and $\delta_2 \ll 1$, what are the approximate maximum passband and stopband approximation errors for $G(e^{jw})$. (5)

- (c) You may have determined in part (b), Tukey's twicing method improves the passband approximation error, but increases the stopband error. Kaiser and Hamming generalized the twicing method so as to improve both the passband and the stopband. They called their approach "sharpening." The simplest sharpening system that improves both passband and stopband is shown in Figure 2. Assume again that the impulse response of the basic system (denoted by frequency response $H(e^{jw})$) is as given in part (a). Repeat part (b) for the system of Figure 2. (5)

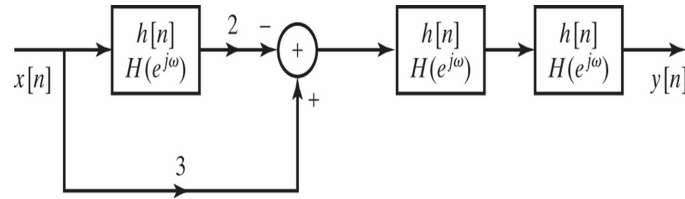


Figure 2: Sharpening approach.